Timelines Test

Instructions

- Show all working. Answers without supporting calculations may not receive full marks.
- Round off your answers to at most four decimal places.
- This is closed-book test. You may NOT use your notes.

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1. Lerato invests R18 500 for 4 years. During the first 18 months, the investment earns 10.2% p.a. compounded monthly. For the remaining 30 months, the rate is 8.8% p.a. compounded quarterly. **[6]**

1.1 Calculate the value of the investment after the first 18 months. **[2]**

1.2 Calculate the final value of the investment after 4 years. **[2]**

1.3 Determine the total interest earned over the full 4 years. **[2]**

2. A savings account is opened with R32 000 at an interest rate of 9.6% p.a. compounded monthly. Exactly 14 months after the account is opened, R6 500 is withdrawn. The remaining money stays invested at the same rate until the end of 4 years. **[6]**

2.1 Calculate the balance immediately before the R6 500 withdrawal. **[2]**

2.2 Calculate the amount in the account at the end of 4 years. **[4]**

3. A school leaver deposits R25 000 into an investment account. For the first 2 years, the account earns 7.5% p.a. compounded quarterly. At the end of the 2 years, an additional R8 000 is deposited. The bank then changes the rate to 9.2% p.a. compounded semi-annually for the next 3 years. **[8]**

3.1 Find the value of the original R25 000 after the first 2 years. **[2]**

3.2 Find the total amount in the account immediately after the additional R8 000 deposit. **[2]**

3.3 Calculate the value of the account at the end of the 5-year period. **[4]**

4. A car savings fund starts with R40 000 and earns 8.4% p.a. compounded monthly. After 18 months, R10 000 is withdrawn. Six months later, R5 000 is deposited into the same account. From the date of that R5 000 deposit, the bank changes the rate to 10.5% p.a. compounded quarterly. The account is left untouched until the end of 4 years from the original deposit. **[10]**

4.1 Calculate the balance just before the R10 000 withdrawal. **[2]**

4.2 Calculate the balance immediately after the R10 000 withdrawal. **[1]**

4.3 Calculate the balance immediately after the R5 000 deposit and interest earned up to that point. **[3]**

4.4 Calculate the final balance at the end of the 4-year period. **[4]**

Goodluck Hlumelo